

Data Communication

Content 09: Cellular Communication

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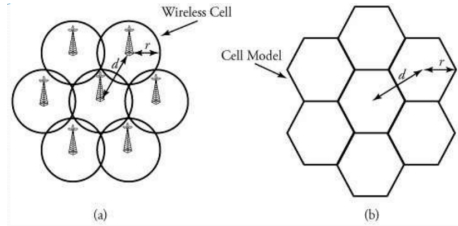


Cellular Communication

- **Definition:** Cellular communication divides a large geographic area into smaller regions called **cells**, each served by a base station.
- **Purpose:** To enable **wireless voice and data communication** for mobile users while efficiently reusing frequencies.
- **How it works:**
 - Mobile users communicate with the nearest base station.
 - Calls or data sessions are transferred (handoff) as users move between cells.
 - Frequency reuse increases system capacity.
- **Evolution:**
Analog 1G → digital 2G (GSM/CDMA, TDMA & CDMA access) → 3G → 4G/5G (OFDMA-based access – outside our scope here).
- Useful video: https://www.youtube.com/watch?v=1JZG9x_VOwA



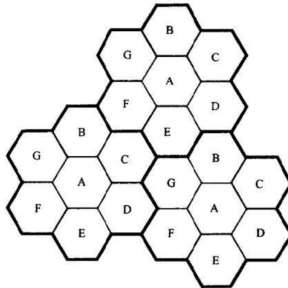
Cell



- A **cell** is the basic geographical coverage area in a cellular network, controlled by a **Base Station (BS)**.
- Hexagons are used because they tile a region with the *fewest* cells (vs. squares/triangles) and closely approximate the circular coverage of an omnidirectional antenna.
- Each cell is assigned its own set of frequencies to limit interference with neighboring cells.

Frequency Reuse

- Adjacent cells are assigned *completely different* channel groups; a base station's antenna is designed to cover only its own cell.
- Because coverage is limited to the cell boundary, the *same* group of channels can be reused in another cell far enough away that interference stays tolerable. This is **Frequency Reuse**.
- The set of cells that together use the *complete* set of available channels exactly once is called a **Cluster**; the cluster size is denoted N .



Frequency Reuse: The Numbers

Cluster size:

$$N = i^2 + ij + j^2, \quad i, j \in \mathbb{Z}^+$$

Common valid values: $N = 1, 3, 4, 7, 9, 12, 13, \dots$

Reuse distance (center-to-center distance between two cells using the same channel group), where R is the cell radius:

$$D = R\sqrt{3N}$$

Worked example: Take $i = 2, j = 1 \Rightarrow N = 4 + 2 + 1 = 7$, and $R = 1$ km.

$$D = \sqrt{3 \times 7} = \sqrt{21} \approx 4.58 \text{ km}$$

A larger N means channels are reused farther apart (less interference) but *fewer channels per cell*; a smaller N gives more channels per cell but more co-channel interference. This trade-off is the core design problem of cellular planning.



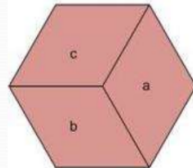
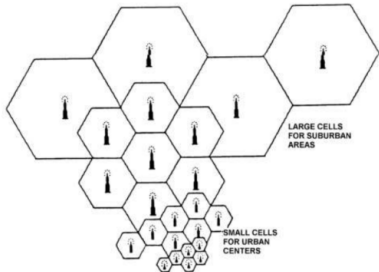
Handoff

- When a mobile moves into a different cell while a call is in progress, the network automatically transfers the call to a new channel belonging to the new base station. This is called **handoff**.
- **Types:**
 - **Hard Handoff:** “Break before make”
 - Existing link is broken before a new one is made.
 - Risk of call drop but simpler implementation.
 - Used with TDMA-based access (e.g., GSM).
 - **Soft Handoff:** “Make before break”
 - New link is established before breaking the old one.
 - More reliable, especially for fast-moving users.
 - Used with CDMA-based access.
- Ensures seamless communication without call drops.

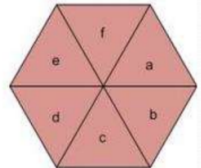


Capacity Improvement Techniques

- As user demand increases, capacity must be improved *without* increasing spectrum.
- **Cell Splitting**: divide large cells into smaller ones, each with its own base station and lower transmit power $\Rightarrow$ same channels reused more often per unit area $\Rightarrow$ more total capacity.
- **Sectoring**: replace one omnidirectional antenna with 3 (120°) or 6 (60°) directional antennas per cell $\Rightarrow$ reduces co-channel interference $\Rightarrow$ allows a smaller reuse cluster N .

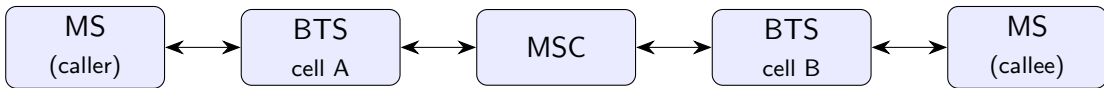


120° sectoring



60° sectoring

General Cellular Architecture



MS (Mobile Station) – the user's handset.

BTS (Base Transceiver Station) – the radio equipment covering one cell.

MSC (Mobile Switching Centre) – routes the call/data between BTSs, and out to other networks (e.g., PSTN, other operators).

This three-tier chain (handset → local base station → central switch) is common to essentially all cellular standards (GSM, CDMA, LTE, 5G).



Thank You

